Model A · Diabetic Retinopathy Grading · Linear Probe

Frozen RETFound-DINOv2 ViT-Large backbone · Linear head only · 50 epochs · 4-class (R0/R1/R2/R3A)

What is Linear Probing?

All ~307 M backbone weights are FROZEN — gradients flow only through a single linear layer (1024 → 4). This measures how much retinopathy information is already encoded in the RETFound-DINOv2 representations without any task-specific adaptation, establishing a clean lower bound before the more expensive full fine-tune.

0.841	0.660	0.867
Best Val AUROC	Macro Sensitivity	Macro Specificity

Training Configuration

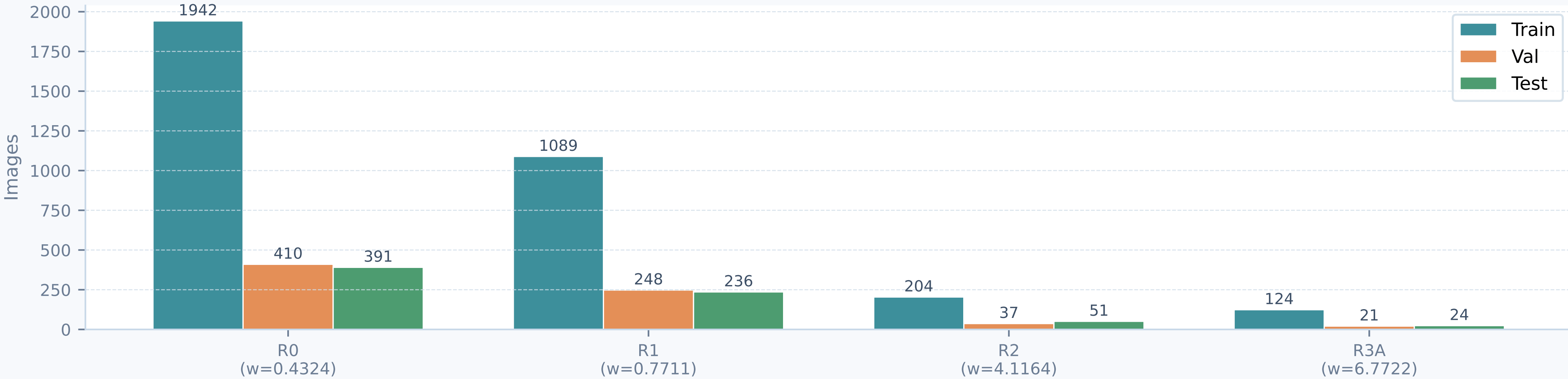
Backbone	RETFound-DINOv2 ViT-Large (~307M params)
Pretrained on	UK Biobank + MEH fundus cohort (736k images)
Backbone status	FROZEN — weights not updated
Head	Linear: 1024 → 4 classes (no hidden layers)
Epochs	50 (best ckpt by val AUROC at epoch 38)
Batch size	64
Input	224 × 224 px colour fundus photo
Optimiser	AdamW (default RETFound settings)
Loss	CrossEntropyLoss with inverse-frequency weights
Class weights	R0=1.00 · R1=1.79 · R2=9.53 · R3A=15.68
Best ckpt metric	Val AUROC → checkpoint-best.pth
Test eval	Deferred (run after fine-tune for direct comparison)

Class Imbalance & Weighting

The dataset is severely skewed — R0 outnumbers R3A by ~16:1. Without correction the model would learn to predict R0 for everything, achieving high accuracy while completely missing sight-threatening proliferative disease.

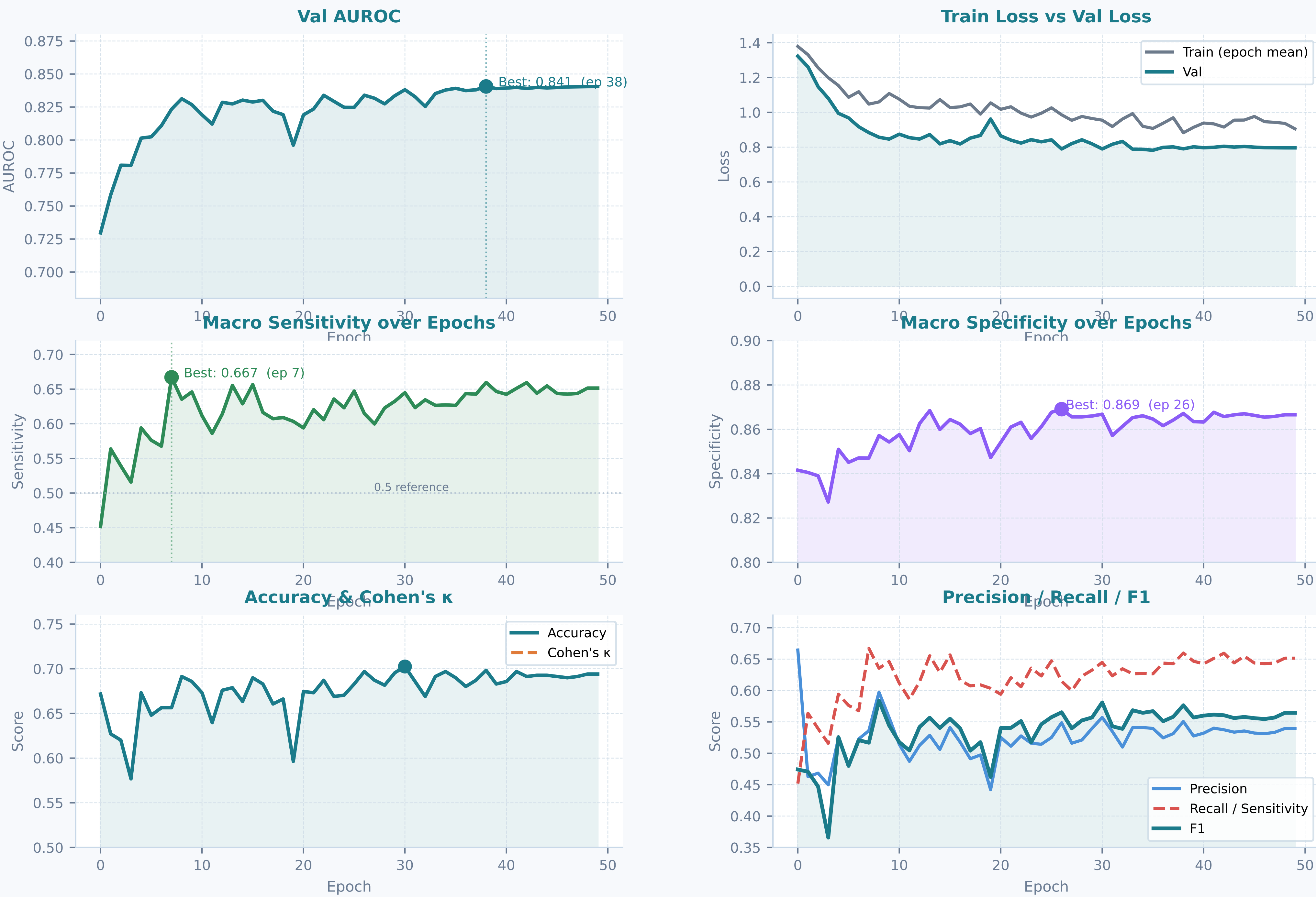
Weight formula: $w_i = N / (n_{classes} \times count_i)$
Scaled so min weight = 1.0 → each class contributes equally to the loss regardless of frequency.

Dataset · Image counts per class and split



Training Dynamics — All Validation Metrics over 50 Epochs

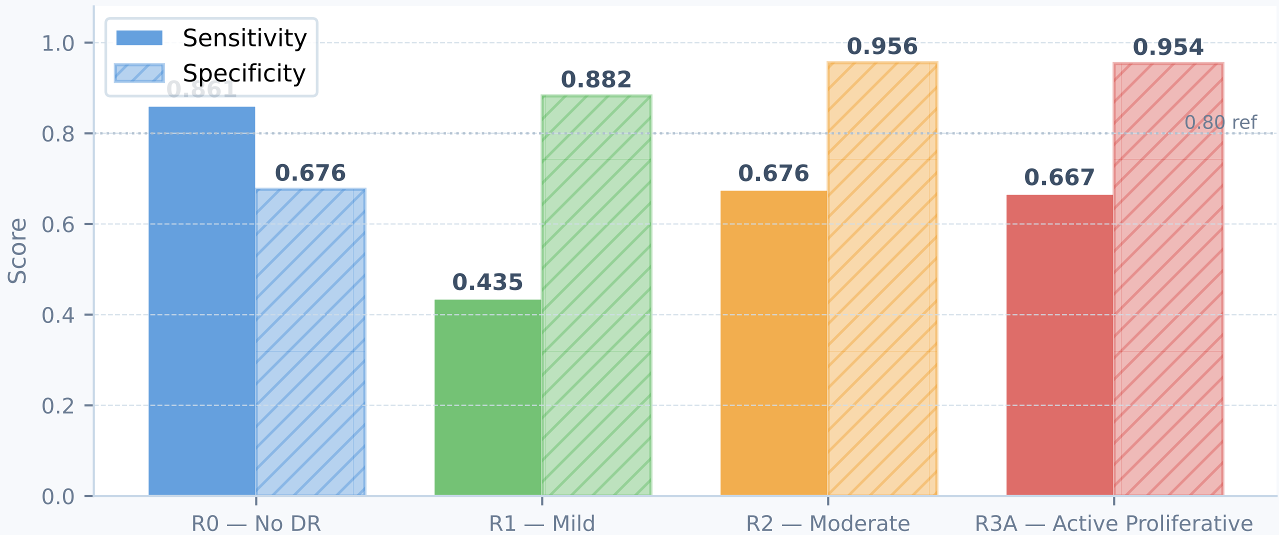
Frozen backbone · only the linear classification head receives gradient updates each epoch



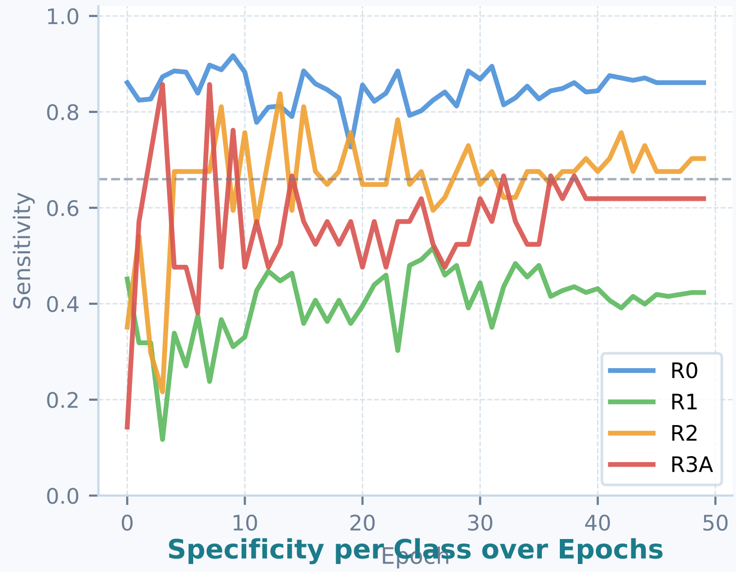
Per-class Sensitivity & Specificity — Linear Probe

Sensitivity = TP/(TP+FN) · Specificity = TN/(TN+FP) · OvR decomposition at best-AUROC checkpoint

Sensitivity & Specificity per Class (best checkpoint, epoch 38)



Sensitivity per Class over Epochs



Specificity per Class over Epochs

Metric Summary at Best Checkpoint (epoch 38, AUROC 0.841)

Class	Sensitivity	Specificity	Clinical Note
R0 — No DR	0.861	0.676	High sensitivity: model catches most healthy eyes
R1 — Mild	0.435	0.882	Weakest sensitivity — mild DR hard to separate from R0 linearly
R2 — Moderate	0.676	0.956	Moderate recall; high specificity avoids over-alerting
R3A — Active Prolif.	0.667	0.954	Good sensitivity despite only 124 training images
Macro average	**0.660**	**0.867**	**Frozen backbone; full fine-tune expected to improve all classes**

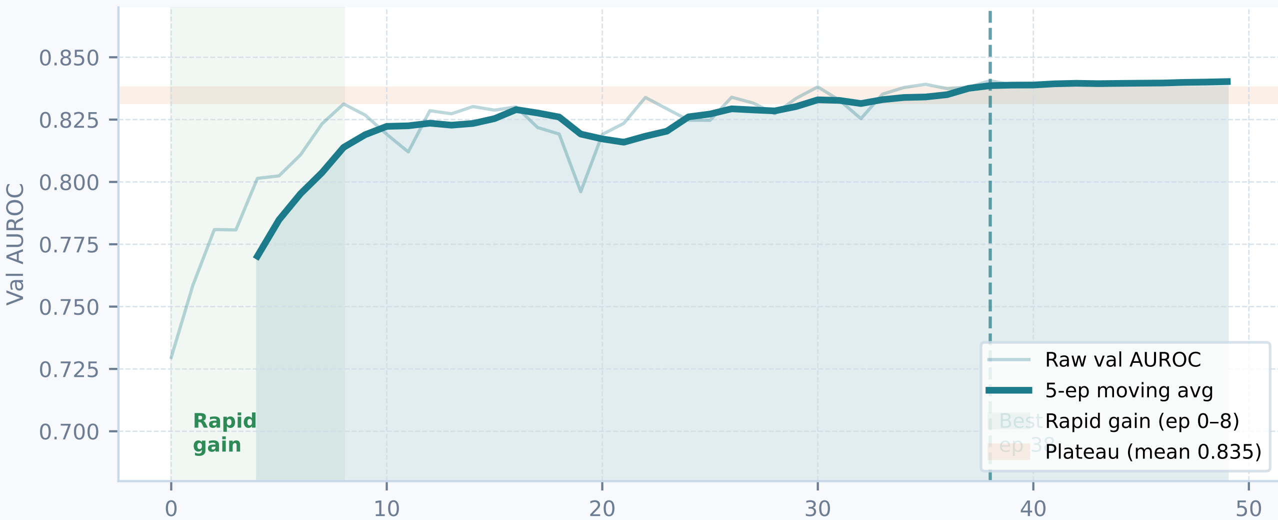
Clinical Interpretation of Sensitivity vs Specificity

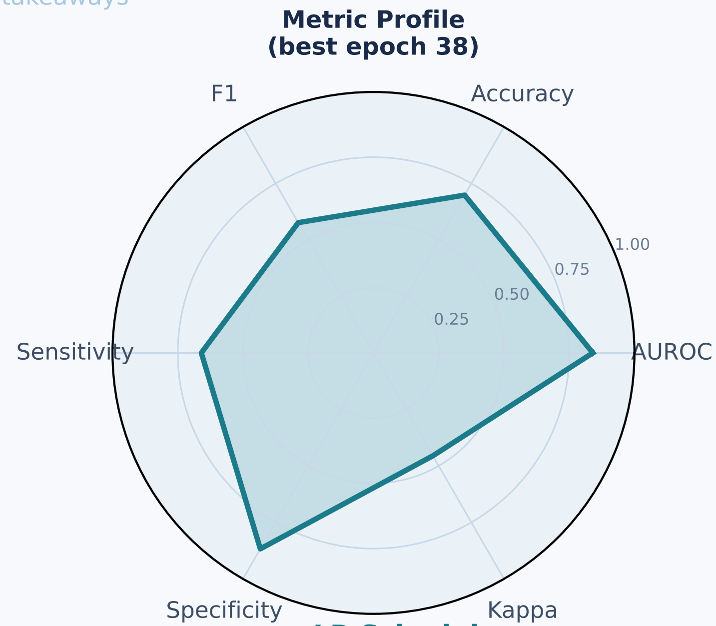
- **Sensitivity (recall):** Low sensitivity = missed cases. Clinically dangerous for R2/R3A— undetected proliferative disease can lead to preventable blindness.
- **Specificity:** Low specificity = false alarms. R0 specificity of 0.68 means 32% of healthy eyes are flagged— acceptable as a screening tool but increases unnecessary referrals.
- **R1 bottleneck:** R1 sensitivity of 0.44 is the weakest point. Mild DR is a borderline grade that overlaps with both R0 and R2 in feature space— a linear head cannot form a curved boundary.

Convergence Analysis & Summary

Understanding training behaviour · plateau detection · key takeaways

AUROC Convergence (raw + 5-epoch moving average)

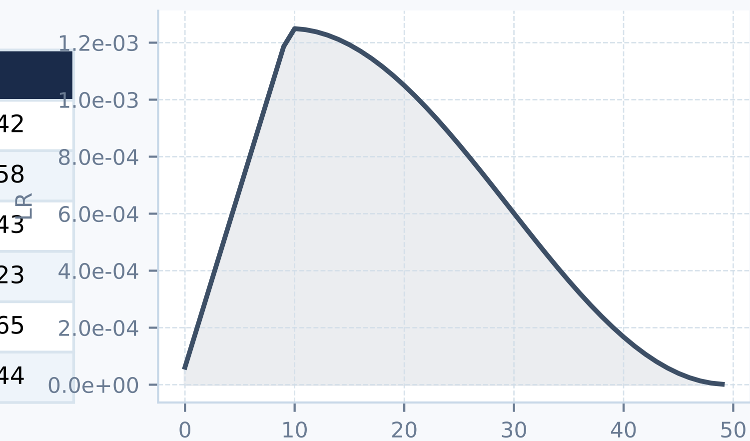




Key Metrics at Selected Epochs

Epoch	AUROC	Accuracy	Kappa	Sensitivity	Specificity	F1
Ep 0	0.7296	0.6718	0.3660	0.4517	0.8416	0.4742
Ep 4	0.8014	0.6732	0.3980	0.5940	0.8510	0.5258
Ep 9	0.8268	0.6858	0.4169	0.6460	0.8543	0.5443
Ep 19	0.7961	0.5964	0.3381	0.6035	0.8473	0.4623
Ep 38 ★	0.8406	0.6983	0.4553	0.6597	0.8672	0.5765
Ep 49	0.8405	0.6941	0.4501	0.6515	0.8666	0.5644

LR Schedule



Key Observations

- AUROC climbed rapidly in the first 8 epochs (0.730 → 0.831), confirming RETFound features already carry substantial retinopathy structure without any adaptation.
- The model plateaued around AUROC 0.836–0.841 from epoch 20 onwards; the linear head saturated and additional training offered negligible gains.
- Macro specificity (0.867) substantially exceeds macro sensitivity (0.660) — the model is conservative: it rarely raises a false alarm but does miss cases, especially R1.
- R1 sensitivity (0.436) is the critical weakness: mild DR sits in a feature-space region that cannot be cleanly separated from R0 by a single hyperplane.
- R2 and R3A achieve reasonable sensitivity (0.676 / 0.667) despite being rare classes — the class weights successfully force the model to attend to these grades.
- This baseline establishes AUROC 0.841 as the frozen-feature lower bound; full fine-tuning of all backbone layers is expected to push significantly higher.